Lab Goals

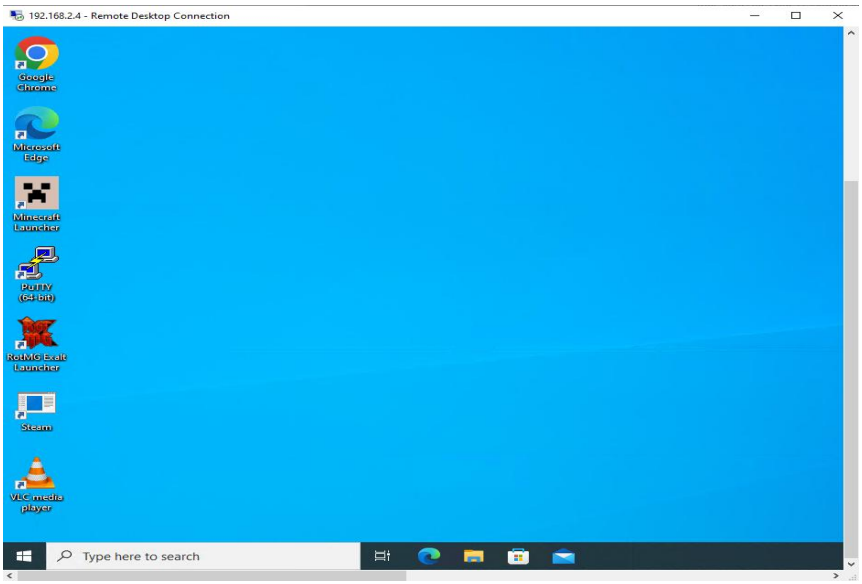
The purpose of this lab is to create a site-to-site VPN encrypted using IPSEC with the encryption method of our choice. We should also configure tunnel-monitor then use Wireshark to determine if the VPN is up or not. We should be able to remote desktop between both computers through the VPN Tunnel.

Background Information

A VPN is a virtual private network, connecting two points together. The main purpose of a VPN is to secure data transmitted between two points, bypass third party tracking/monitoring, and allows secure data transmission when running on public networks. When sending data through public networks, the data sent between the two devices can be intercepted by the intermediaries that the data must go through to reach the destination. The two main types of VPNs are Site to Site VPNs and Point to Point VPNs. Site to site VPNs are used to connect several different networks together such as multiple corporate networks. This allows tight network security between these points, where all the data transmitted across can be encrypted and provides security to the networks where private IP addresses are hidden from anyone outside the network. The way these VPNs are configured can also be incredibly flexible, with the ability to add more sites to the VPN whenever convenient, even on public WIFI networks. A Point to point will create a virtual private network between two endpoints. P2p networks are extremely secure with only limited encryption needed to make sure the data isn't easily exposed. This is because while Site to Site VPNs can run over public networks, p2p networks will run only on this virtual network that you have created. This means that it is extremely difficult for attackers to infiltrate this virtual network you have created and only devices you have added will exist on it. However, every location you want to add a P2P VPN on will require a dedicated device such as a firewall or router to translate this virtual network data into your network. Site to Site VPNs generally can be more flexible and cheaper to make and P2P VPNs provide maximum security. In this lab, we are creating a P2P VPN and as such, the rest of this information will be regarding P2P VPNs. Palo Alto VPNs can be created with IPSec and GRE. IPSEC will always provide more data integrity, authentication, and encryption. IPSEC is most beneficial to provide maximum security inside of your VPN tunnel. This is most beneficial in cases where data security is much more important than flexibility of your networks. GRE will be much more flexible in allowing a large number of networks to access an endpoint. This is incredibly useful in cases such as WANs, especially over the internet. It is important to know that GRE and IPSEC can be mixed to create a tunnel with both IPSEC's security and GRE's flexibility. Palo Alto firewalls have additional VPN capabilities. The main capability that will coincide with these VPNs are the network zones feature. To allow a VPN into your Palo alto firewall's network, you need to first create a network zone for the VPN, because without it the firewall will automatically block that traffic. To create an IPSEC VPN, an IKE Gateway is required to enter the tunnel. IKE Gateways are used to terminate and initialize VPN connections between the networks. They are configured with an encryption method which will be how the data between the two networks are encrypted and what they are encrypted with. For this VPN to work, a SOHO network must be configured on both sides of the network with the devices you want to secure directly connected to the firewall.

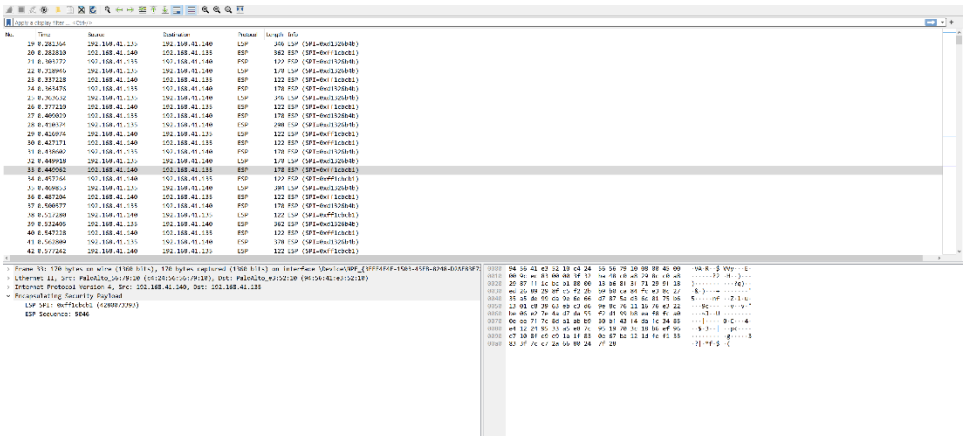
Lab Summary

To set up our Site-to-Site VPN on our Pa-220 firewall, we created an IKE gateway to negotiate the tunnel's adjacency between our two sites using both an IKE Encryption method and the IPSec encryption method itself. This serves as the end points of both tunnels. Next, we set up the IPSec tunnel itself to transport and encrypt the data sent between both points. To send data between both points, we set up a VPN zone which is the zone all tunnel traffic will run on. However, even if our tunnel is up, we are still unable to ping between both interfaces because the security policies were blocking us from pinging the other side. Thus, we entered Policies > Security and created a policy that allowed traffic to be sent and received from the tunnel interface. It's very helpful to see why your tunnels are failing and when, so we set up tunnel monitors. After enabling both interfaces, then our tunnel interfaces would be up. After this, we entered lusrmgr.msc and created a profile on both computers. After adding them both to groups, we turned on remote desktop in windows setting and then attempted to remotely access the other PC in the "remote desktop connection" app. Entering the password we configured in lusrmgr.msc, we were able to access the other desktop remotely.



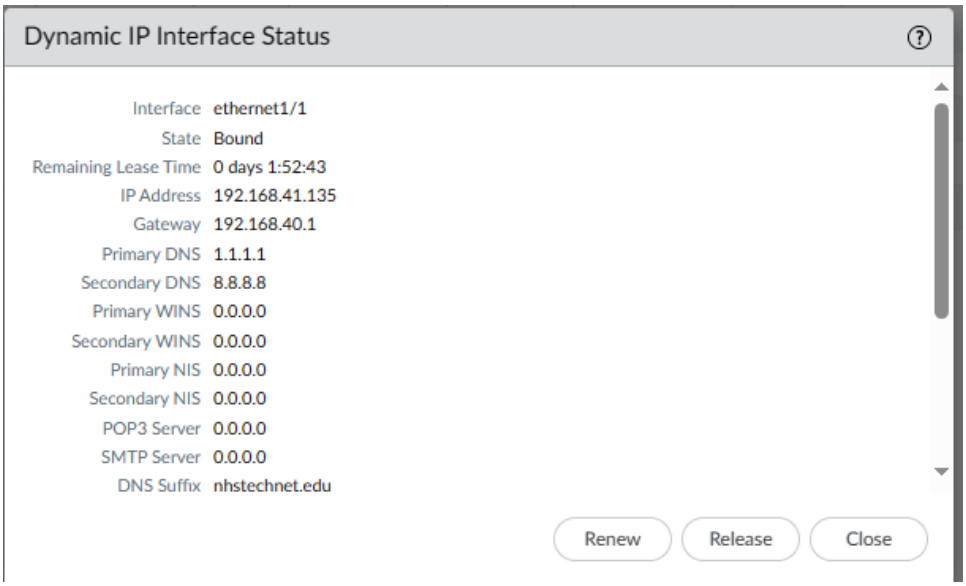
	NAME	STATUS	TYPE	IKE Gateway/Satellite INTERFACE	LOCAL IP	PEER ADDRESS	STATUS	Tunnel Interface INTERFACE	VIRTUAL ROUTER	VIRTUAL SYSTEM	SECURITY ZONE	STATUS	COMMENT
	ipsec-tunnel	Tunnel Info	Auto Key	ethernet1/1		192.168.41.140	IKE Info	tunnel.20	default (Show Routes)	vsys1	VPN		

After this, pinged both computers from across the IPSec tunnel and IPSec traff ic between both ends should be encrypted with ESP



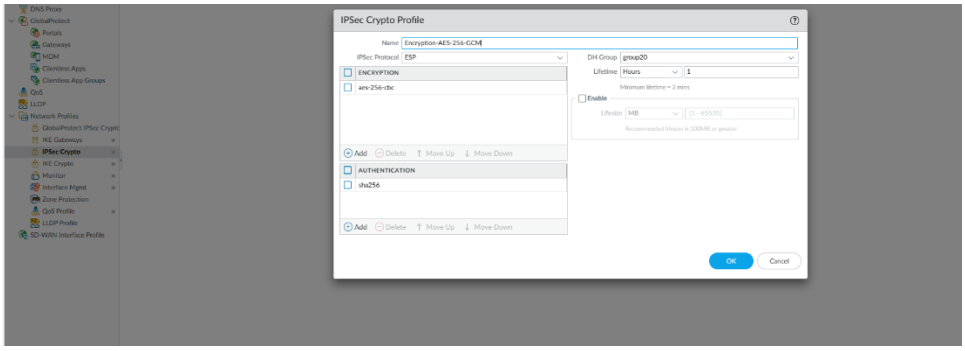
Lab Commands

1.Allocate an IP so DHCP does not change the IP address of the outbound interface each time the firewall reboots. Make sure both sides of the firewall have an IP allocated to them on DHCP.

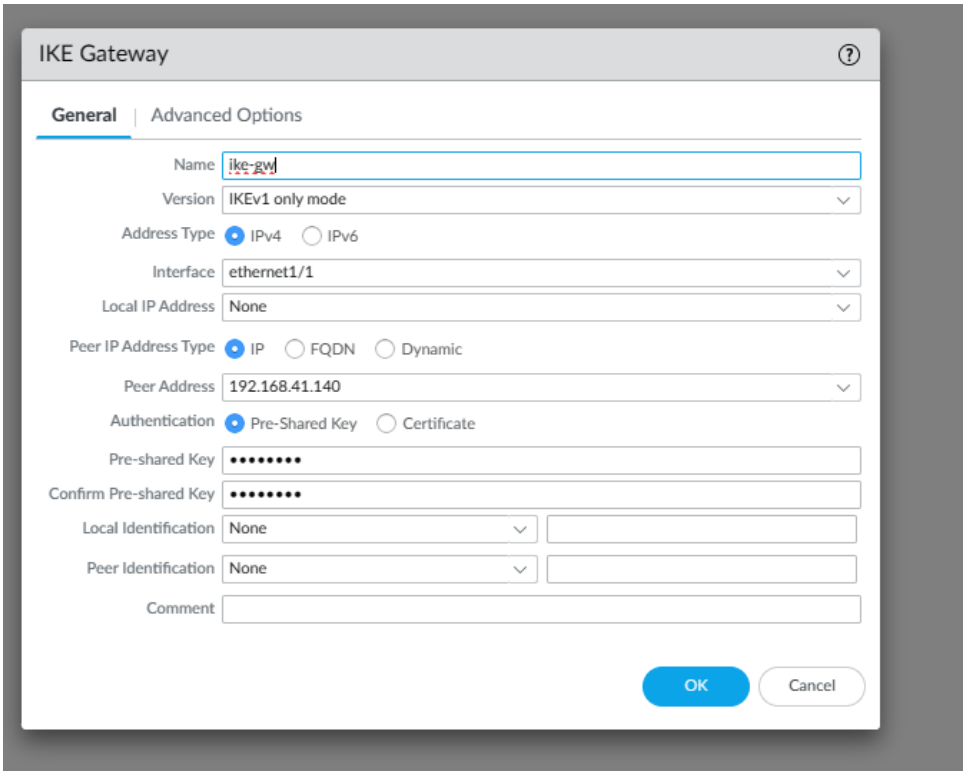


2.Create an IKE Crypto profile in Network > IKE Crypto with a group number and encryption/auth method. Make sure these settings are the same for both sides of your VPN tunnel firewalls.

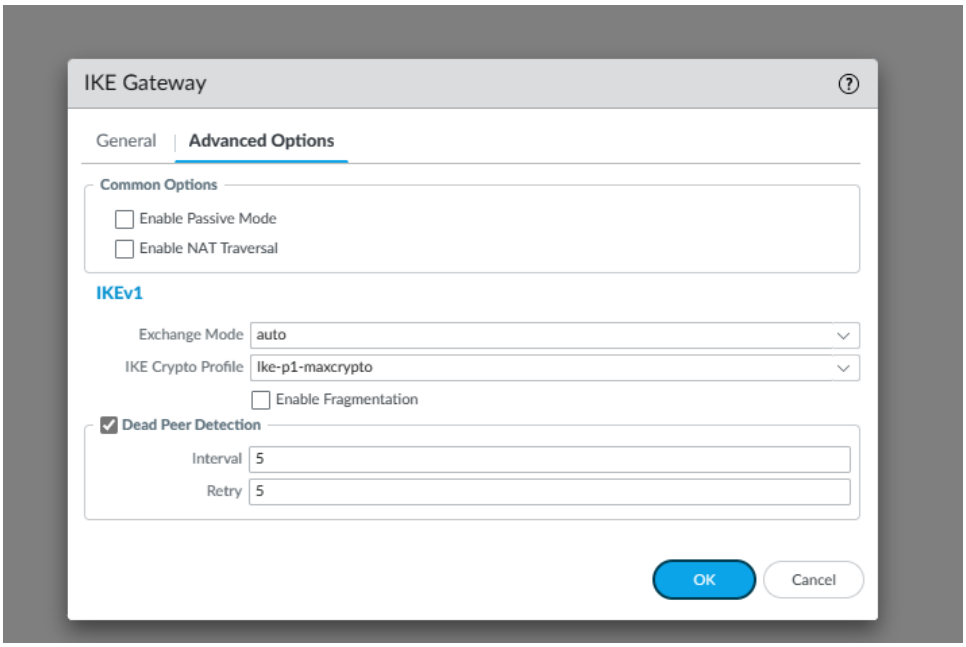
3.Create an IPSEC Crypto profile running ESP Protocol and select your encryption method and group number. Make sure this group number is consistent throughout the configuration process on both Firewalls.



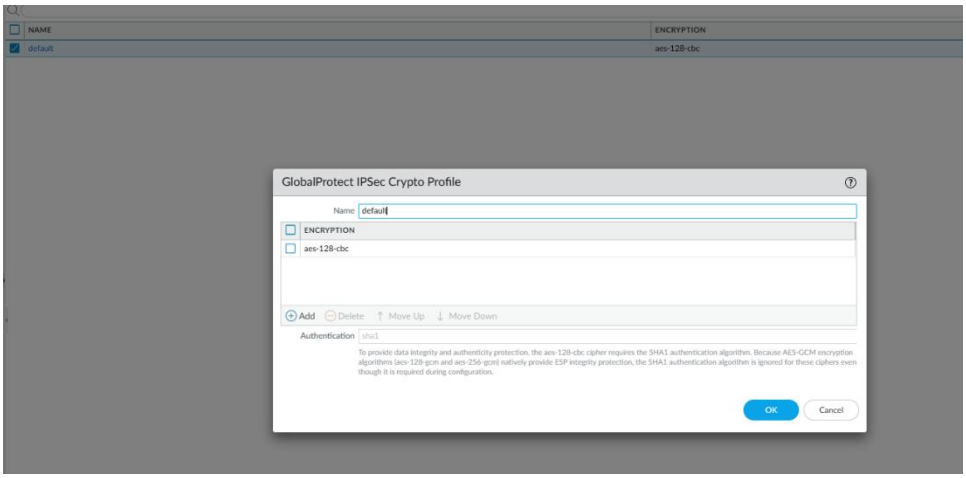
4. Create an IKE Gateway in Network > IKE Gateway. Configure a name and version. For this firewall, we used IKEv1 only mode and thus our configuration will be different from one that uses a different version. Configure the interface to be the outbound interface that is connected towards the destination firewall. If you have a static IP configured instead of DHCP, you will select the local IP address and use the outbound interface IP. Insert the address of your peer's outbound interface which should be on the same network as you and select a pre-shared key. Make sure both sides have the same pre-shared key.



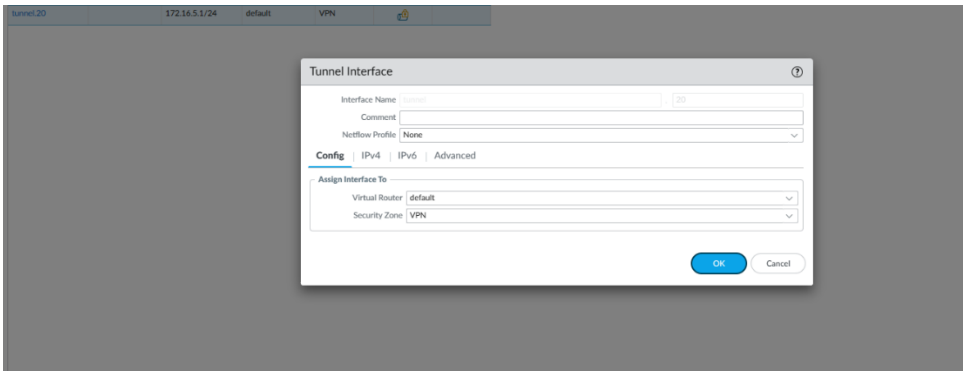
5.Select Advanced Options and set your IKE Crypto Profile to the Profile you configured in step 2. Make sure the right profile is selected if you have multiple and is the same on both firewalls.

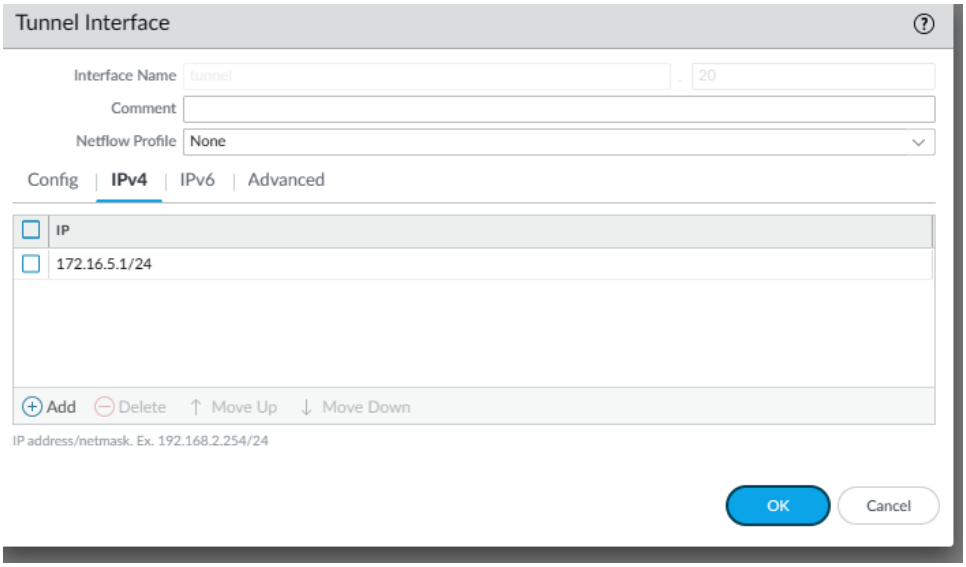


6.Enter Network > GlobalProtect IPsec Crypto and select an encryption method.

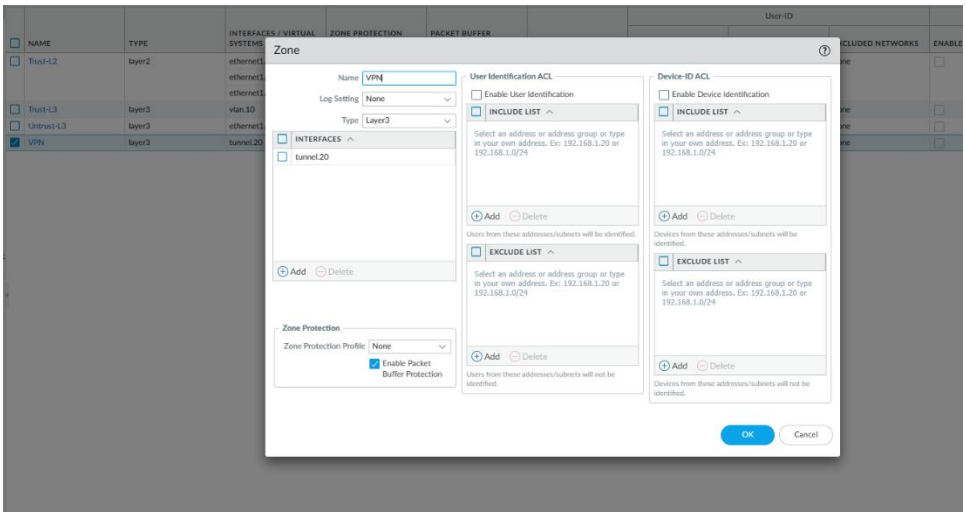


7.Create a tunnel interface and set and select the virtual router and the security zone. Next, set a private IP address in the network you want the tunnel to run on. This IP should not be on your own private network or public network.

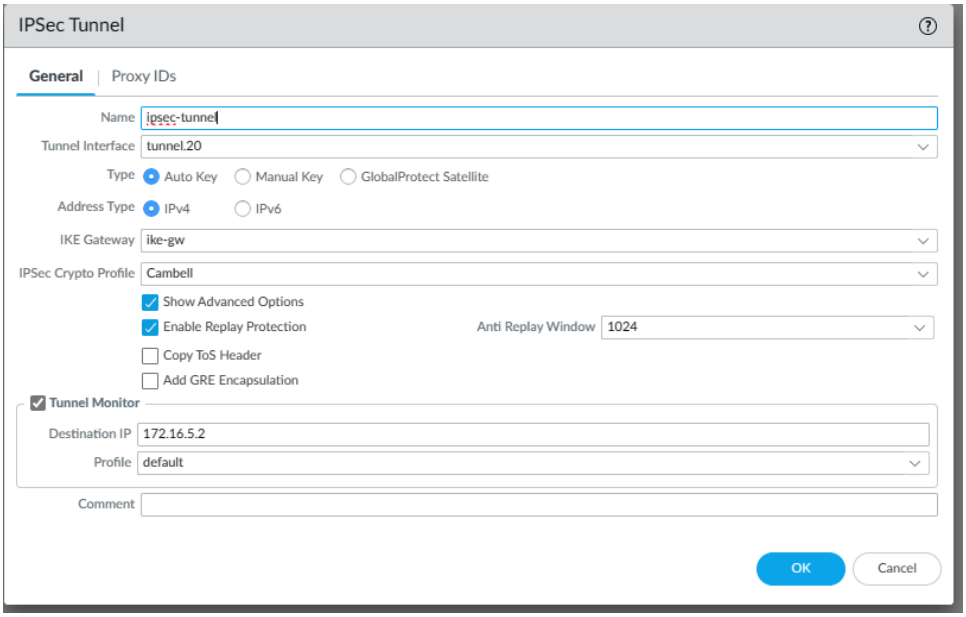




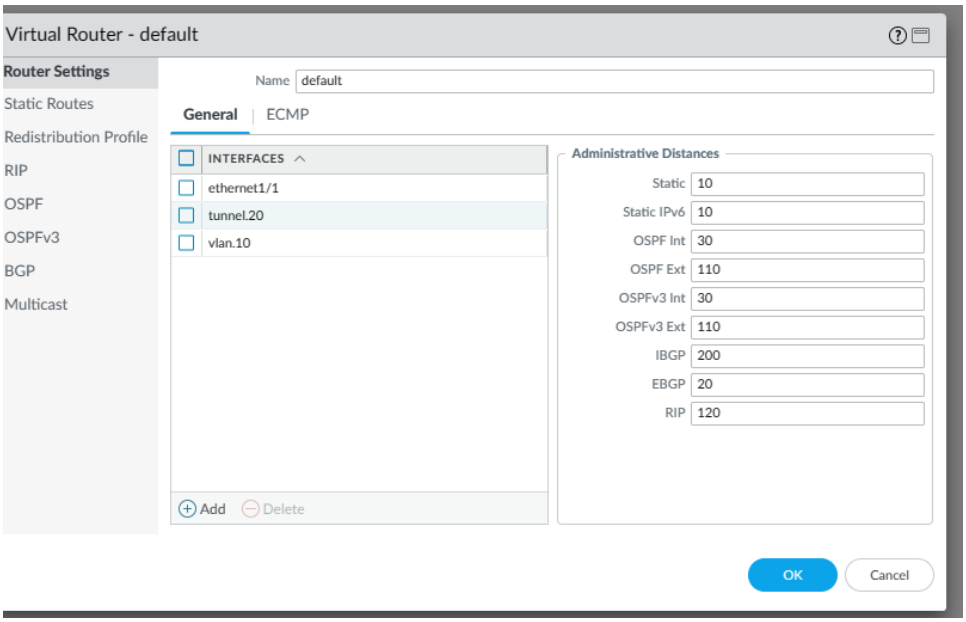
8.Create a security zone that your VPN will run on. Set the tunnel interface to the interface you created in step 8.



9.Enter Network > interfaces > IPSEC Tunnel Name your IPSEC Tunnel, set your tunnel interface to the interface you created in step 8, select your desired key type, and select an address type of IPV4. Select the IKE Gateway that was configured in step 4 and the crypto profile you created in step 5. Enable replay protection. If you want to enable tunnel monitoring, set a destination IP in the network you configured in step 8.



10.Enter Network > Virtual Router and add your tunnel interface into the Virtual router. Enter static routes and add a static route to the destination private network and select the next hop address.



Virtual Router - Static Route - IPv4

Name: Route
Destination: 192.168.1.0/24
Interface: tunnel.20
Next Hop: IP Address
192.168.1.254
Admin Distance: 10 - 240
Metric: 10
Route Table: Unicast

Path Monitoring

Failure Condition: Any / All
Preemptive Hold Time (min): 2

NAME	ENABLE	SOURCE IP	DESTINATION IP	PING INTERVAL(SEC)	PING COUNT

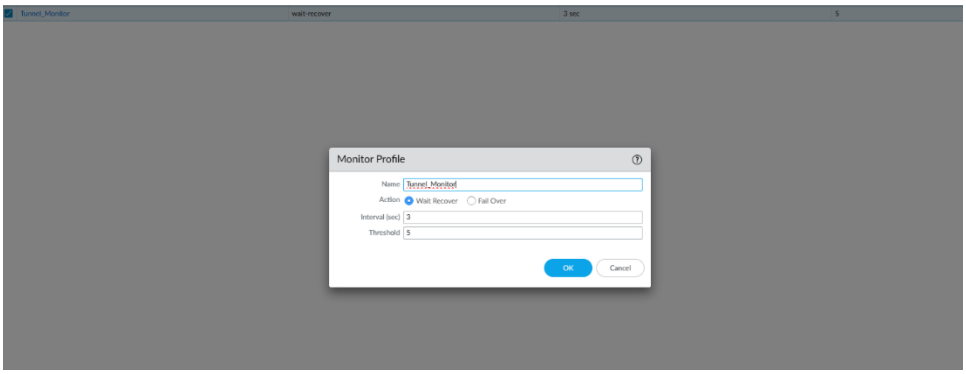
Add Delete

OK Cancel

11.Enter Policies > Security and create two security policy rules. One to enter the destination network and one to come back from the destination network. Make one from your private network zone and the destination to the VPN zone and make the other policy allow traffic from the VPN zone to your private network. Enter “Actions” and select allow.

2	to-tunnel	none	universal	Trust-L3	any	any	any	VPN	any	any	any	application-...	Allow	none		98	2023-1
3	from-tunnel	none	universal	VPN	any	any	any	Trust-L3	any	any	any	application-...	Allow	none		135	2023-1

12.Enter Network > Monitor and create a monitor profile.

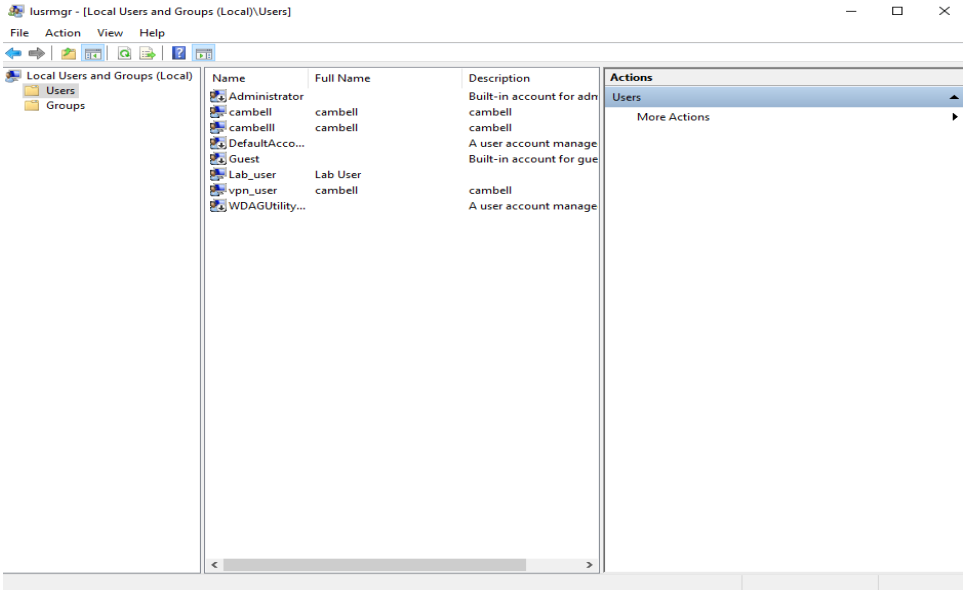


13.Enter Network IKE Gateway and IPSec Tunnel and then enable both interfaces in the bottom left hand corner

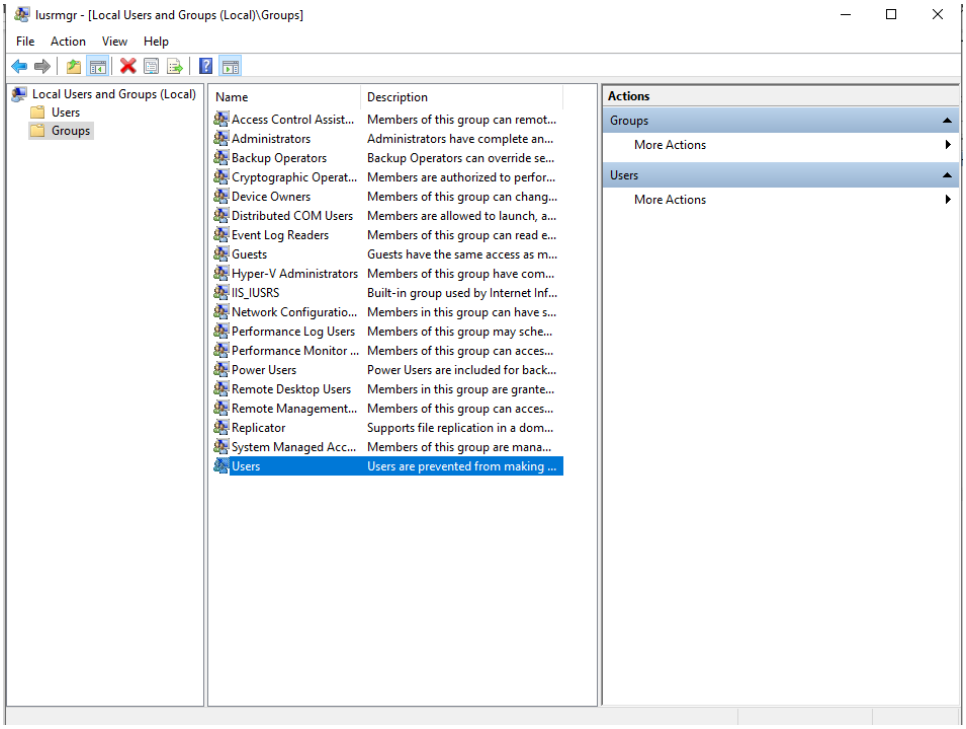
			Local Address		Peer ID		Local ID				IKE Advanced Options						
NAME	PEER ADDRESS	INTERFACE	IP	ID	TYPE	ID	TYPE	VERSION	MODE	PASSIVE MODE	NAT TRAVERSAL	CRYPTO PROFILE	DPD	LIVENESS	COMMENT
ike-gw	192.168.41.140	ethernet1/1						ikev1	auto			ike-p1-manycrypto	enabled(default/...		

Add Delete Enable Disable PDF/CSV

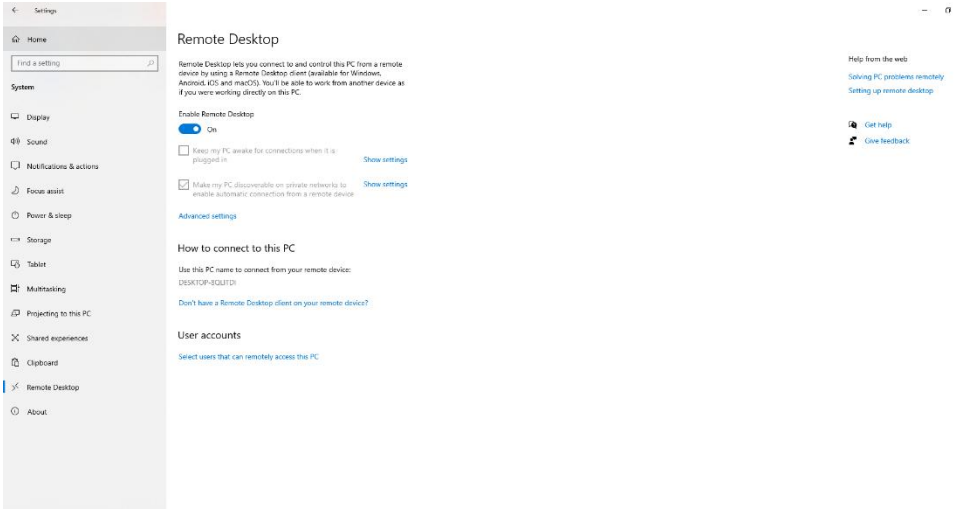
14. On both PCs, Enter lusrmgr.msc and create a user and a password.



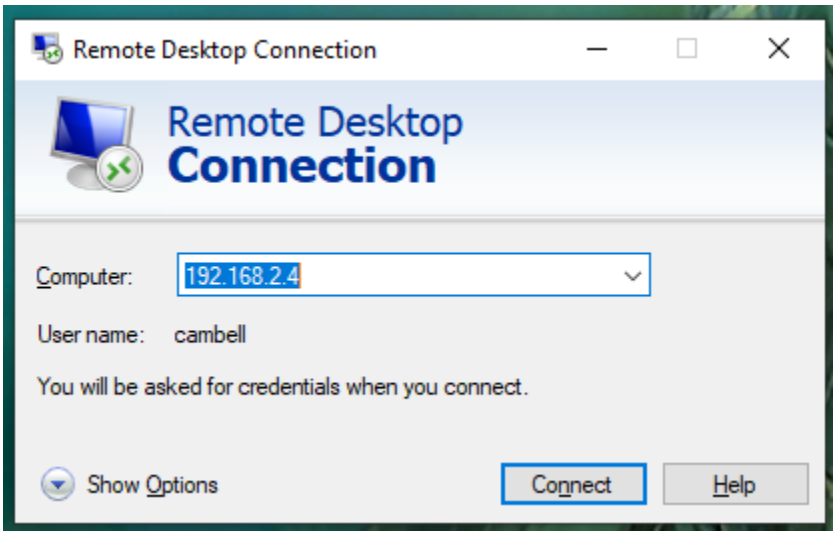
15.In groups, open the folder called ‘users’ and add the profile you create in step 13.



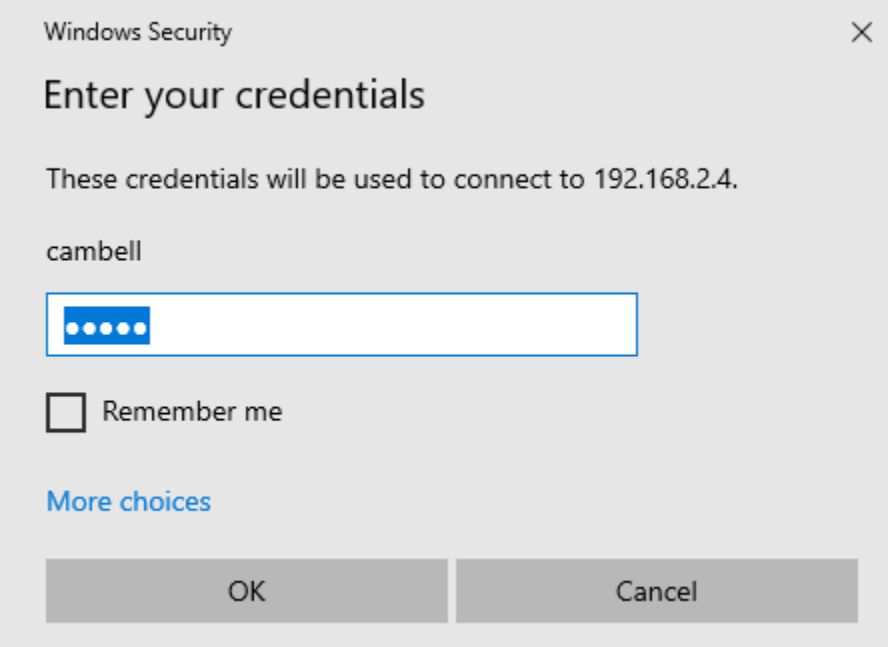
16.On both pcs, open settings and enable remote desktop.



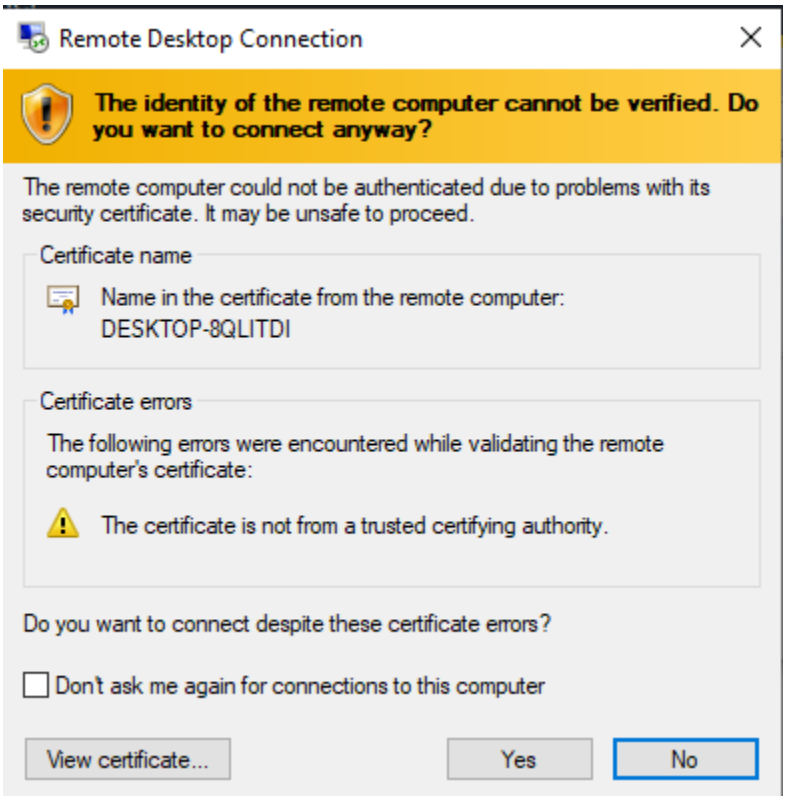
17.From one endpoint, open “remote desktop manager” and enter the private IP address of the other endpoint.



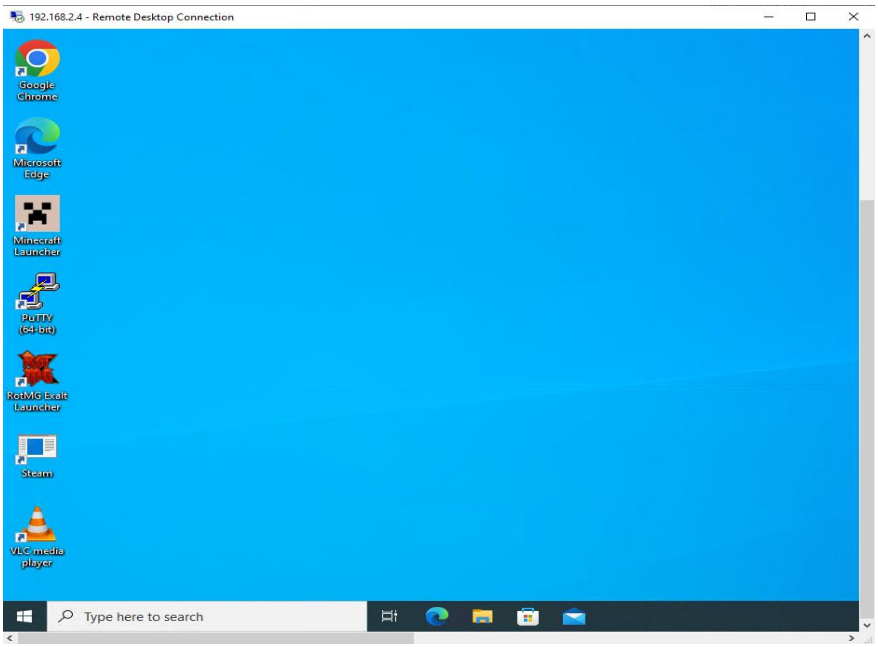
18. Enter the password you created in step 13.



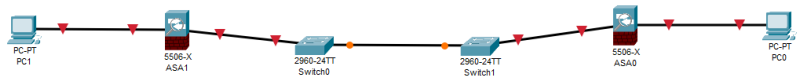
19.Confirm that you have a remote desktop connection and select “yes”.



20.When a prompt appears on the computer that is being remote desktoped into, confirming if you want to allow an external user to remotely access your computer, select “yes”.



Network diagram



Configurations

Network

Interfaces

Tunnel.20

Virtual Router: Default

Security Zone: VPN

IPv4 Address: 172.16.5.1/24

Management Profile: None

Zones

VPN

Interfaces: Tunnel.20

Type: Layer 3

Log Setting: None

Zone Protection Profile: None

Virtual Routers

Default

Interfaces: Tunnel.20

Static Route:

Name: Route

Destination: 192.168.1.0/24

Interface: tunnel.20

Next Hop: IP Address

IP address: 192.168.1.254

Metric: 10

Route Table: Unicast

IPSec Tunnel

IPsec-Tunnel

Tunnel Interface: Tunnel.20

Type: Auto Key

Address Type: IPv4.

IKE Gateway: ike-gw

IPSec tunnel: Encryption-AES-256-GCM

Enabled Replay Protection (advanced option)

Tunnel Monitor: On

Destination IP: 172.16.5.2

Profile: Default

IKE Gateway

Ike-gw

IKEv1 only mode

Address Type: IPv4

Interface: Ethernet1/1

Local IP: None

Peer IP: 192.168.41.140

Preshared Key: 12345678

Local identification: none

Peer identification: none

Advanced:

IKE Crypto Profile: Ike-p1-maxcrypto

IPSec Crypto

Ike-p1-maxcrypto

IPSec protocol: ESP

Encryption: AES-256-CBC

Sha256

DHGroup: Group20

Lifetime: 1 hour.

IKE Crypto

Suite-B-GCM-256

Aes-256-cbc

Sha 256

Group 20

Monitor

Tunnel-Monitor

Action: Wait Recover

Interval: 3 sec

Threshold: 5 sec

Policies

Security

To-tunnel

Name: To-tunnel

Rule type: Universal

Source: Trust-L3

Destination: VPN

Actions: Allow

Profile Type: None

From-tunnel

Name: From-tunnel

Rule Type: Universal

Source: VPN

Destination: Trust-L3

Actions: Allow

Profile Type: None

Problems

Rushing through the lab

At first, we rushed through the lab and attempted to quickly complete it, but our configurations would not work because some of our commands were not matched correctly due to moving past it too fast.

Volatile DHCP address

Every time school closed, the racks would get powered off and our DHCP address would change tomorrow during bootup. As a result, we were unable to configure a local interface IP and destination IP because those were always changing. Fixed by allocating a DHCP address to firewalls.

Palo Alto Website has incomplete commands

The Palo Alto website instructions give a general outline of what you are supposed to do when configuring tunneling, but it doesn't give you all the instructions. Creating to and from tunnels in our security policies was not listed in the palo alto website but was necessary to ensure that traffic can reach the destination network. Similarly, proxy ids were not specified.

Various ways to install commands

We were given numerous options on how we wanted to configure the VPN. Like we could have used IKEv2 as well. We were quickly confused by the Palo Alto website's instructions on how to configure each version, because it seemed like each page had different commands than the others.

Conclusion

A point-to-point VPN running on palo alto can be very reliable and secure in terms of keeping data safe. However, when presented on larger scale networks, it is unrealistic to attempt to scale this VPN. A point-to-point VPN should only be used in small networks where data security is paramount. It is quite complicated to configure a p2p VPN and as such, each settings configuration should be documented carefully to make sure that both sides have the same configuration.

Sign off:

IPSEC VPN Tunnel Lab

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Period 5

Mr. Mason